ASSOCIATED DETAIL SPECIFICATION

RESISTORS, FIXED, FILM, HIGH VOLTAGE, STYLES RHV10, RHV11, RHV12, RHV13, RHV14, RHV15, RHV16, RHV17, RHV18, RHV19, RHV20, RHV21

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for style RHV10, RHV11, RHV12, RHV13, RHV14, RHV15, RHV16, RHV17, RHV18, RHV19, RHV20, and RHV21 resistors.

1.2 Military part number. Resistors covered by this specification shall be identified by a military part number which shall consist of the basic number of this specification and a coded number. The military part number shall be in the following form:



The coded number shall be derived in accordance with MIL-R-49462.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications. The following specifications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-R-39032 - Resistors, Packaging of.

MIL-R-49462 - Resistors, Fixed, Film, High Voltage, General Specification for.

(Copies of the specifications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Laboratory Command, ATTN: SLCET-R-S, Fort Monmouth, NJ 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3. REQUIREMENTS

3.1 Associated detail specifications. The individual item requirements shall be in accordance with MIL-R-49462, and as specified herein.

3.2 Design and construction. The resistor shall be of the design, construction, and physical dimensions specified on figure 1.

3.3 Power rating. The power rating shall be as specified in table I, based on a full load operation at an ambient temperature of 125°C. For operation at temperatures higher than 125°C, derate in accordance with figure 2.

TABLE I. Power rating.

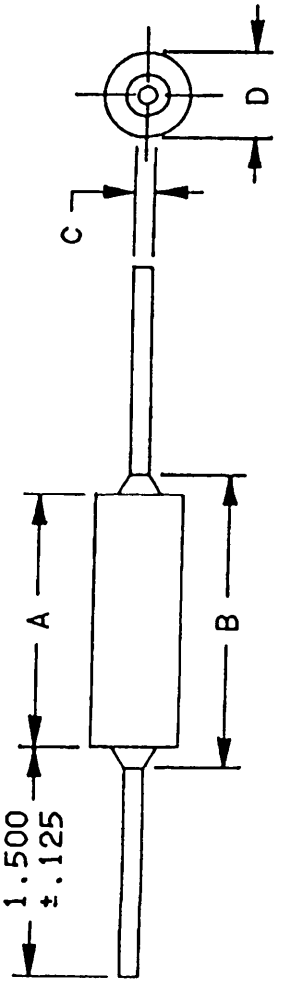
Resistor style	Power rating (watts)	Style
RHV10	0.5	A
RHV11	1.0	B
RHV12	1.5	C
RHV13	2.0	D
RHV14	2.5	E
RHV15	3.0	F
RHV16	5.0	G
RHV17	6.0	H
RHV18	7.5	J
RHV19	8.0	K
RHV20	10.0	M
RHV21	15.0	N

3.4 Resistance. The minimum and maximum nominal resistance values shall be as specified in table II.

TABLE II. Minimum and maximum nominal resistance values.

Resistor style	Resistance value (ohms)		Style
	Minimum	Maximum	
RHV10	1k	1G	A
RHV11	1.2k	2.49G	B
RHV12	1.7k	10G	C
RHV13	2k	10G	D
RHV14	1.5k	7.5G	E
RHV15	2k	10G	F
RHV16	3k	15G	G
RHV17	4k	20G	H
RHV18	600	750M	J
RHV19	800	1000M	K
RHV20	1k	1270M	M
RHV21	1k	2000M	N

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Inches	mm
.002	0.05
.032	0.81
.125	3.18
1.500	38.10

Dimensions								
	RHV10	RHV11	RHV12	RHV13	RHV14	RHV15	RHV16	RHV17
A	0.400 (10.2)	0.775 (19.7)	1.275 (32.4)	1.525 (38.7)	1.550 (39.4)	2.050 (52.1)	3.050 (77.5)	4.050 (102.9)
B max	0.470 (11.9)	0.845 (21.5)	1.345 (34.2)	1.595 (40.5)	1.634 (41.5)	2.134 (54.2)	3.134 (79.6)	4.134 (105.0)
C	0.25 ±.002 (0.64) .051	0.25 ±.002 (0.64) .051	.025 ±.002 (0.64) .051	.025 ±.002 (0.64) .051	.032 ±.002 (.081) .051	.032 ±.002 (.081) .051	.032 ±.002 (.081) .051	.032 ±.002 (.081) .051
D	.135 ±.015 (3.43 ±0.38)	.135 ±.015 (3.43 ±0.38)	.135 ±.015 (3.43 ±0.38)	.135 ±.015 (3.43 ±0.38)	.270 ±.015 (6.86 ±0.38)	.270 ±.015 (6.86 ±0.38)	.270 ±.015 (6.86 ±0.38)	.270 ±.015 (6.86 ±0.38)

FIGURE 1. Resistor, fixed, film, high voltage.

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	RHV18	RHV19	RHV20	RHV21
A	3.125 ±.060 (79.38 ±1.52)	4.000 ±.120 (101.60 ±3.05)	5.000 ±.120 (127.00 ±3.05)	6.000 ±.120 (152.40 ±3.05)
B max	3.265 max (82.93)	4.200 max (106.68)	5.200 max (132.08)	6.200 max (157.48)
C	.040 ±.002 (1.02 ±0.05)	.040 ±.002 (1.02 ±0.05)	.040 ±.002 (1.02 ±0.05)	.040 ±.002 (1.02 ±0.05)
D	.315 ±.040 (8.00 ±0.76)	.315 ±.040 (8.00 ±0.76)	.315 ±.040 (8.00 ±0.76)	.350 ±.040 (8.89 ±1.02)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Millimeters are in parentheses.

FIGURE 1. Resistors, fixed film, high voltage - Continued.

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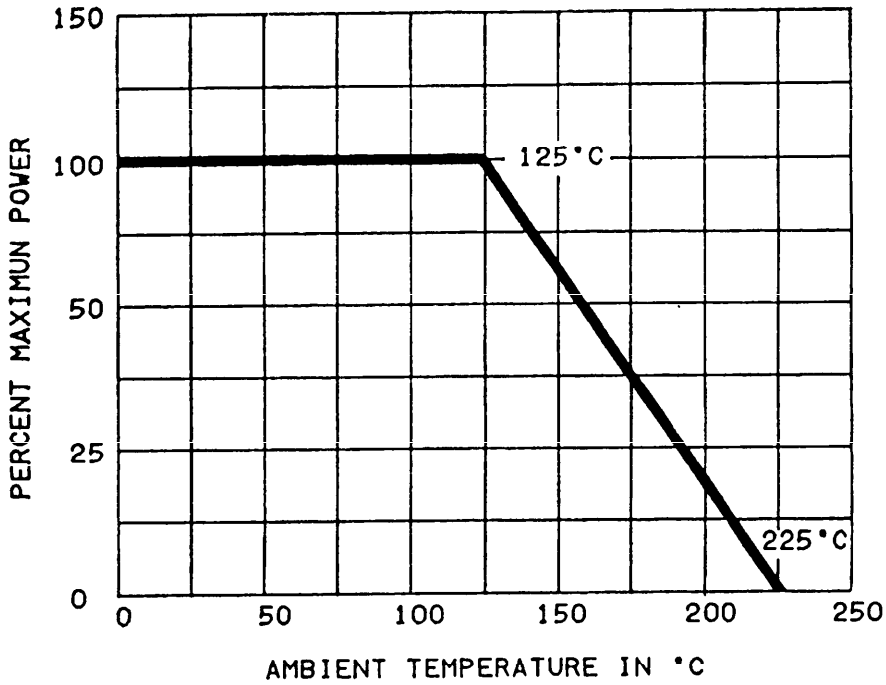


FIGURE 2. Derating curve.

3.5 Maximum voltages. Maximum voltages shall be in accordance with table III

TABLE III. Maximum voltages.

Resistor style	Voltage (volts maximum) (kV)	Style
RHV10	1.5	A
RHV11	2.0	B
RHV12	5.0	C
RHV13	5.0	D
RHV14	11.0	E
RHV15	15.0	F
RHV16	22.5	G
RHV17	30.0	H
RHV18	15.0	J
RHV19	20.0	K
RHV20	25.0	M
RHV21	30.0	N

3.6 Resistance tolerance. The resistance tolerance is identified by a single letter in accordance with table IV.

TABLE IV. Resistance tolerance.

Symbol	Resistance tolerance percent (±)
F	1.0
G	2.0
J	5.0

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-49462.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-R-39032.

6. NOTES

6.1 Intended use. Resistors are intended for use in electronic circuits where high voltages and high resistances are required.

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, MI
Navy - AS, OS
Air Force - 17
DLA - ES

User activities:
Army - AT, AV, ME
Navy - CG, MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-49462/1	RESISTORS, FIXED, FILM, HIGH VOLTAGE, STYLES RHV10 through RHV21

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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